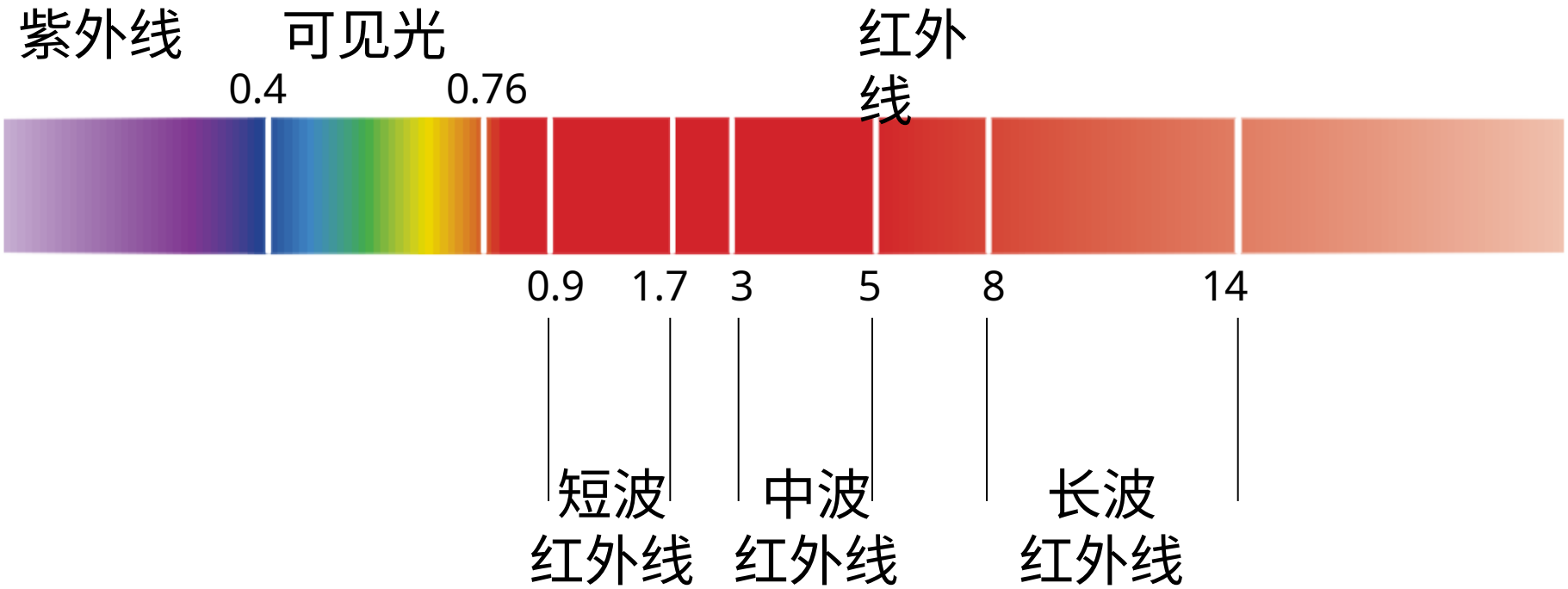
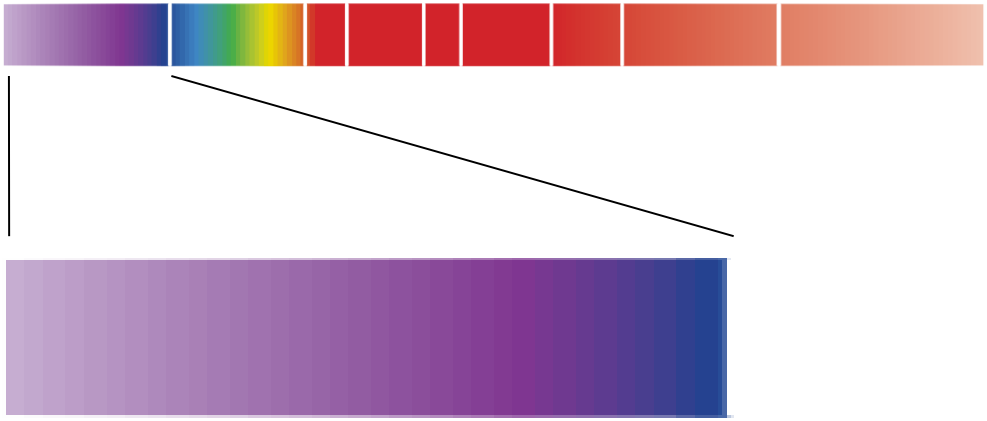
光源

COGNEX

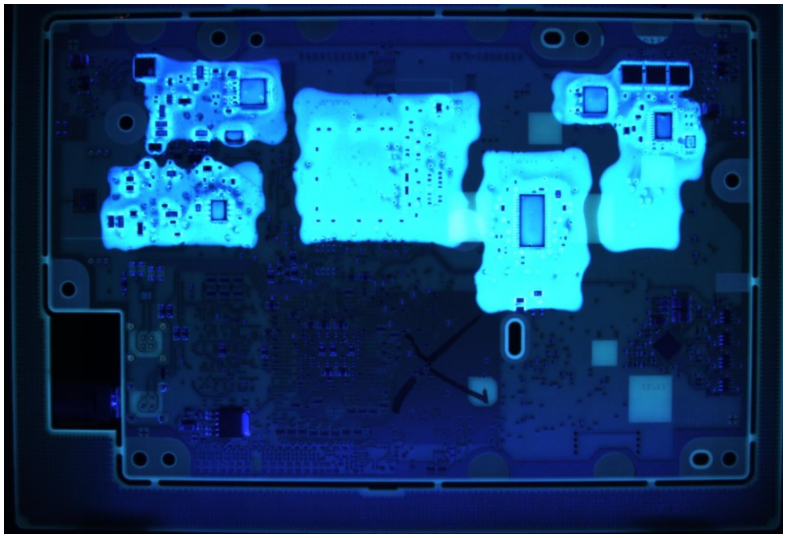
光谱



光谱

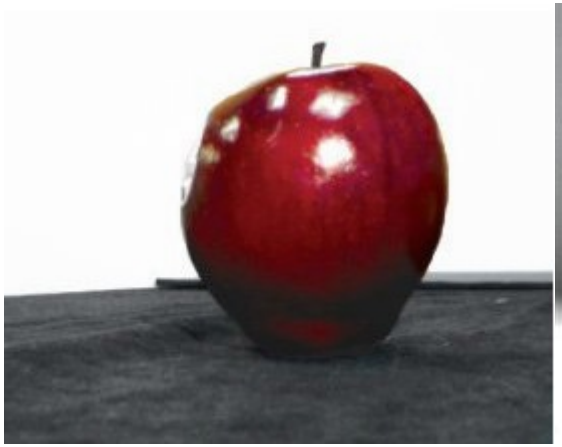


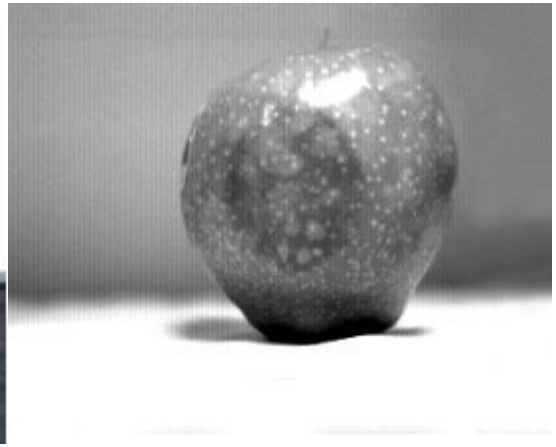
紫外线



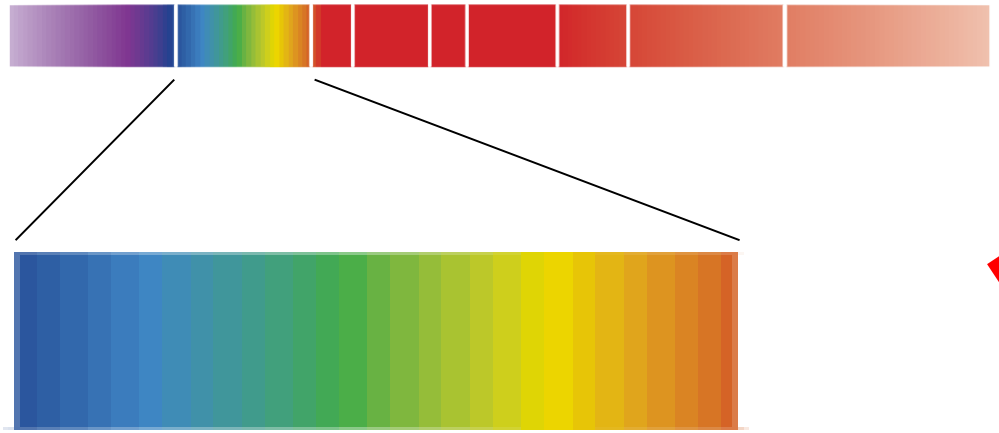
光谱

红外线

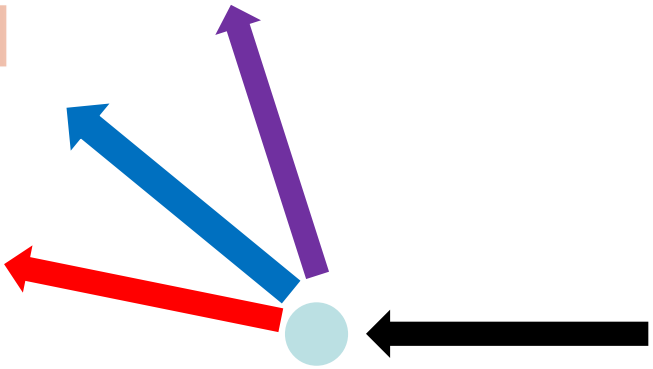




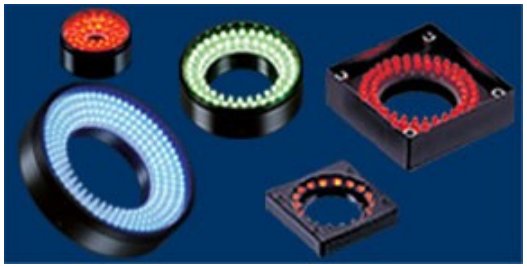
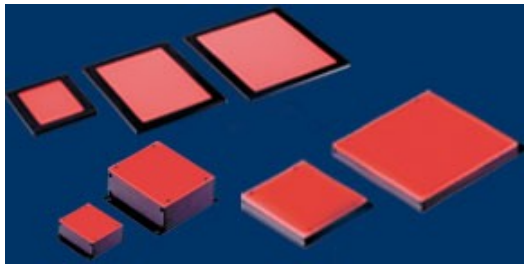
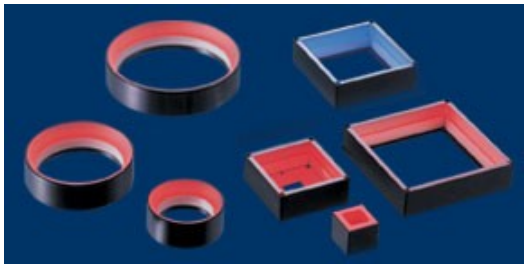
光谱



可见光



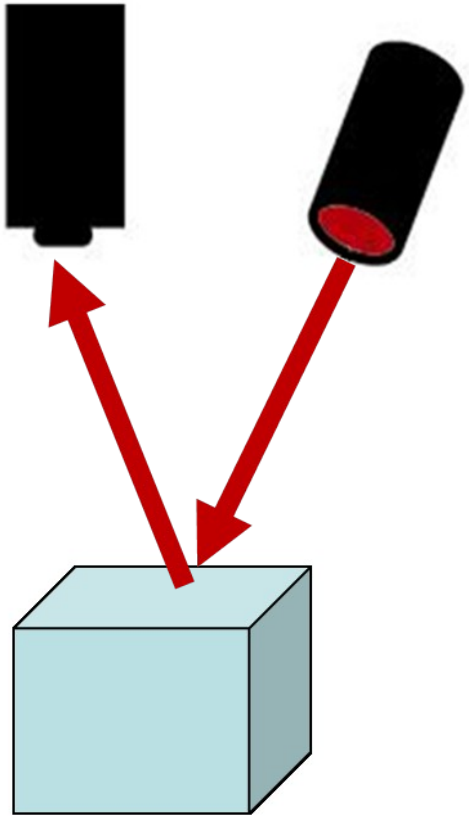
选择光源需要考虑的因素

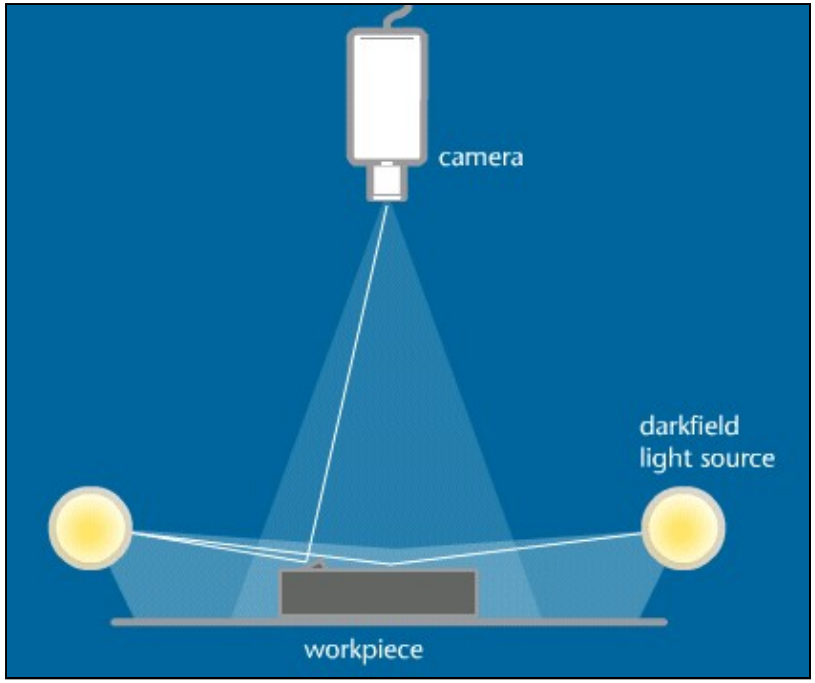
- 表面是镜反射（光滑并有光泽）还是漫反射（粗糙并且暗淡）？
- 表面是否有定向反射比？（以特定的方向反射光）
- 元件的外表在不同颜色光下会发生变化吗？（变深或者变淡）
- 元件的表面是扁平的还是三维的？弯曲的？不规则的？蚀刻的？浮雕？凸起？
- 元件的表面是稳定的还是会随着时间而发生变化？（生锈、氧化、褪色）

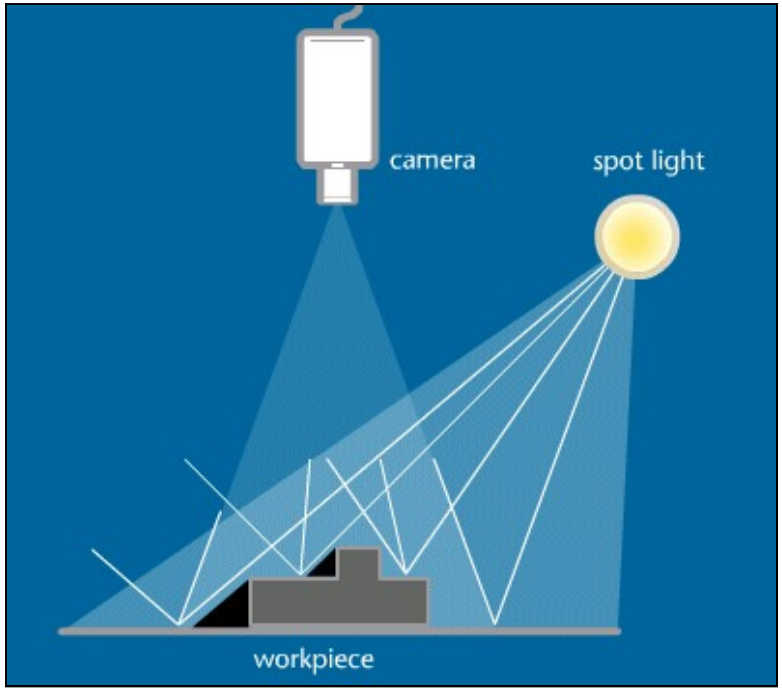
光线和相机

- 当有光线进入相机即可建立图像
  - 亮即有光线直接进入相机
  - 暗即无光线进入相机



光源的选择

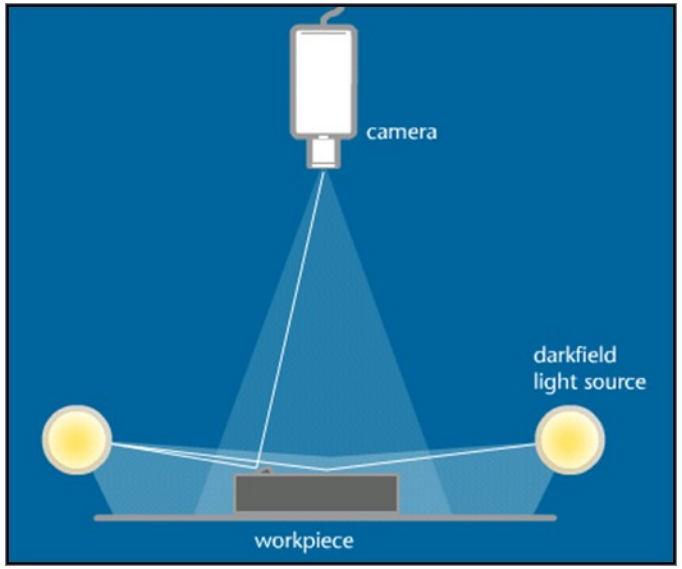


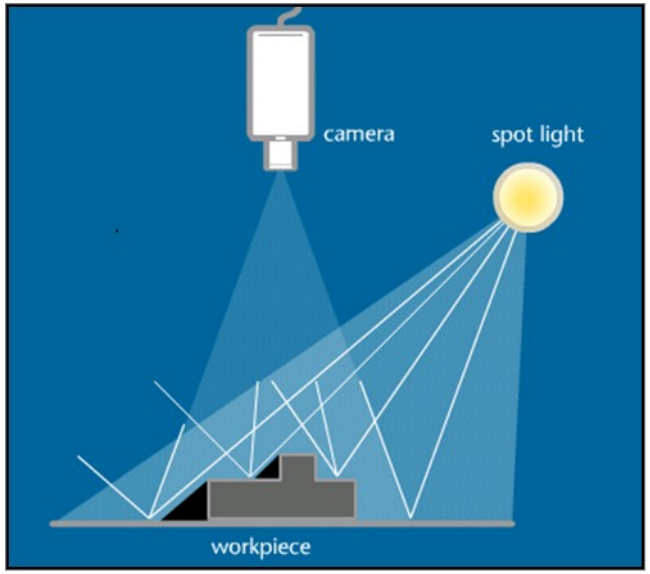


光源可以做成各种形状 .

光源的安装角度和方向直接决定图象的效果 .

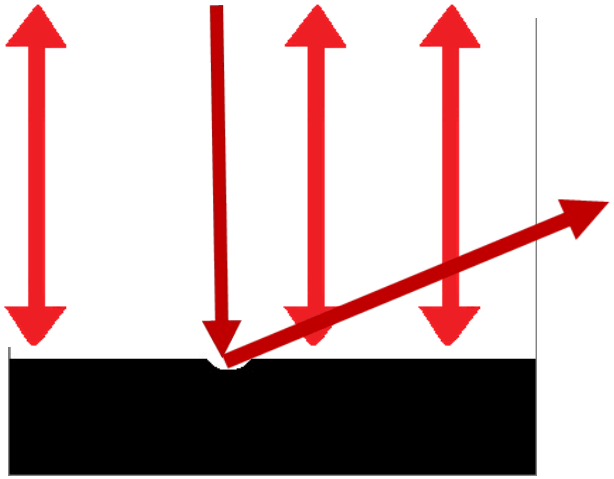
光源的选择

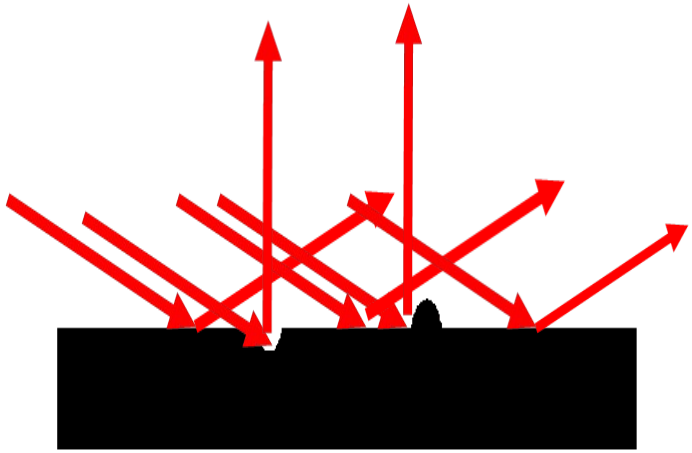




Same Part -
Different Light Position

亮场与暗场





暗视场应用

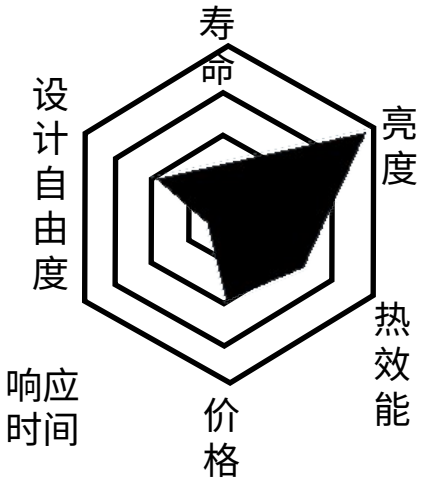
环境光

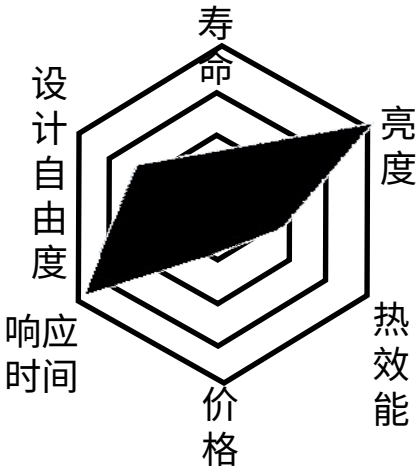
环形光

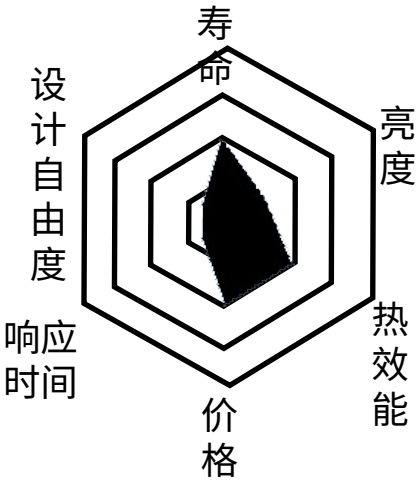
光源的类型

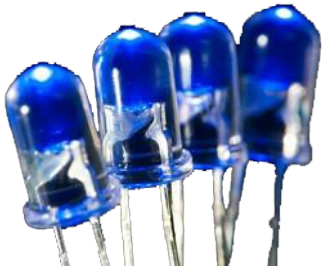
卤素灯
白炽灯
荧光灯
激光
发光二极管（LED ）
氙气灯

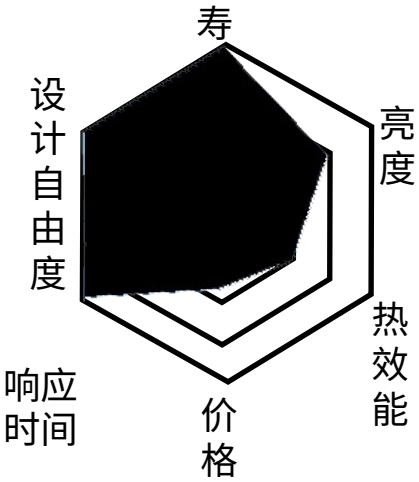
光源种类


寿命
亮度
热效能
价格
响应时间
设计自由度


寿命
亮度
热效能
价格
响应时间
设计自由度

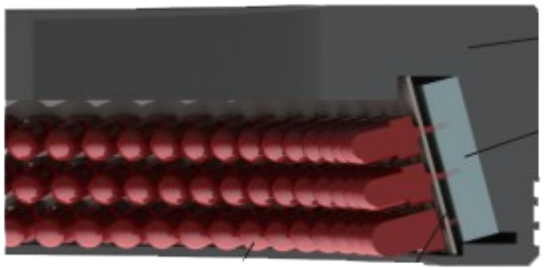

寿命
亮度
热效能
价格
响应时间
设计自由度



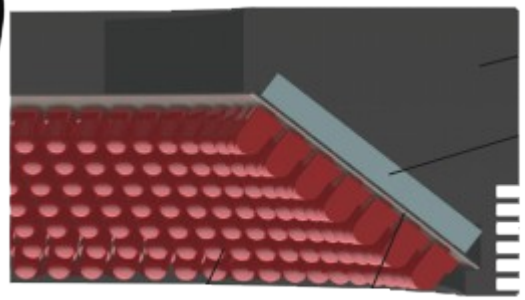

寿命
亮度
热效能
价格
响应时间
设计自由度

环光

0° 环光

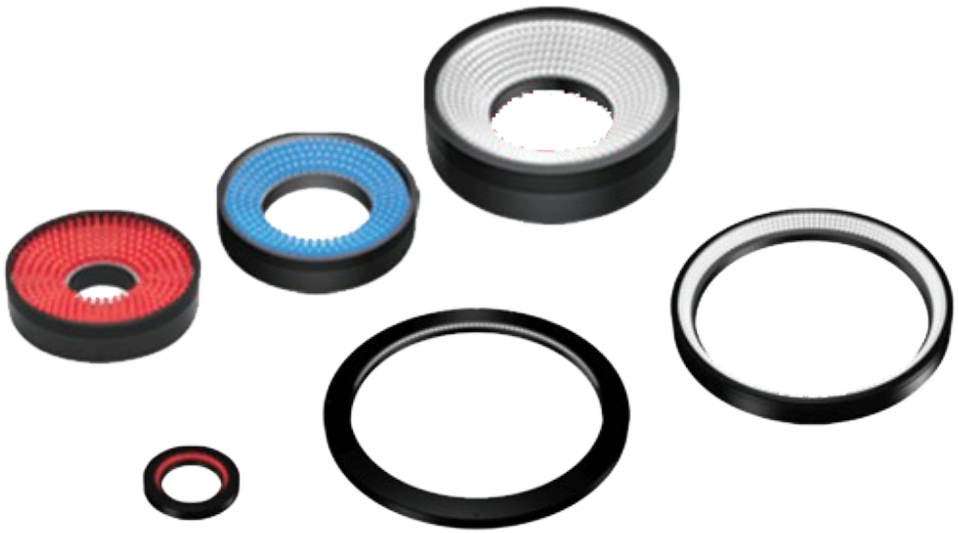


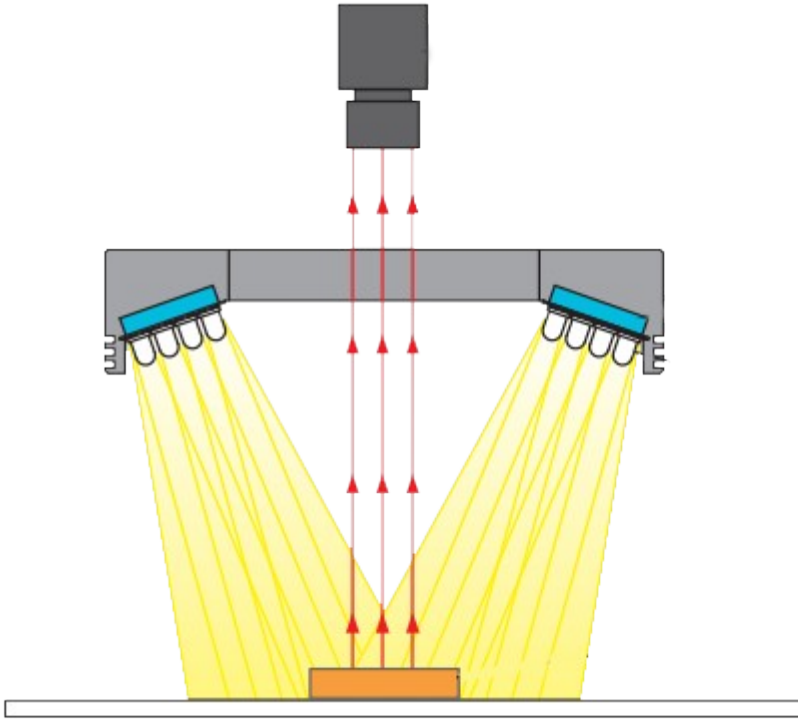
30° 环光



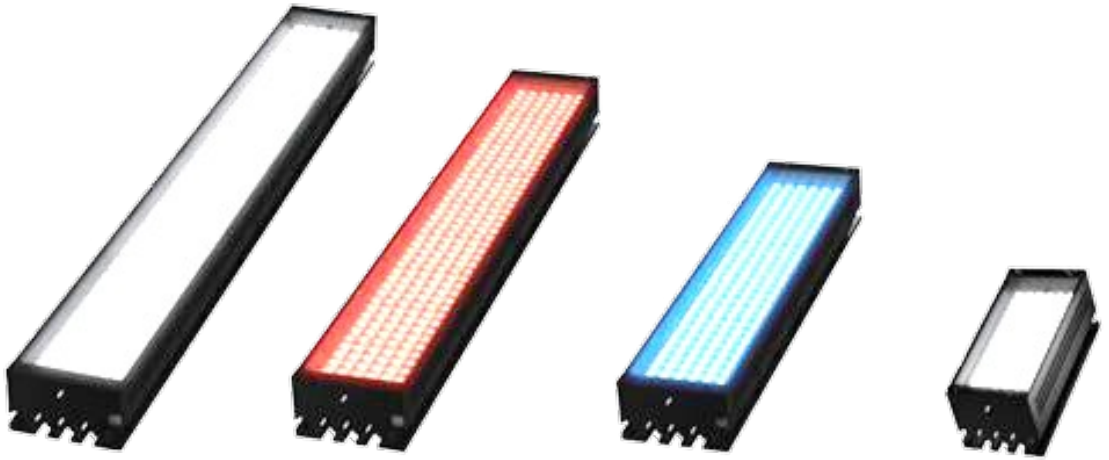
60° 环光

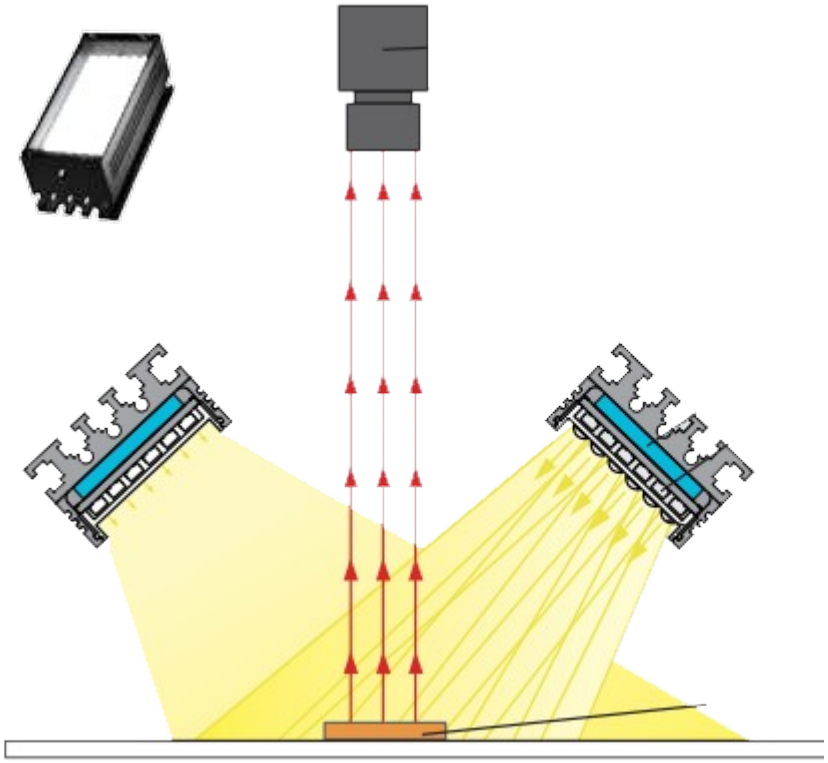
环光



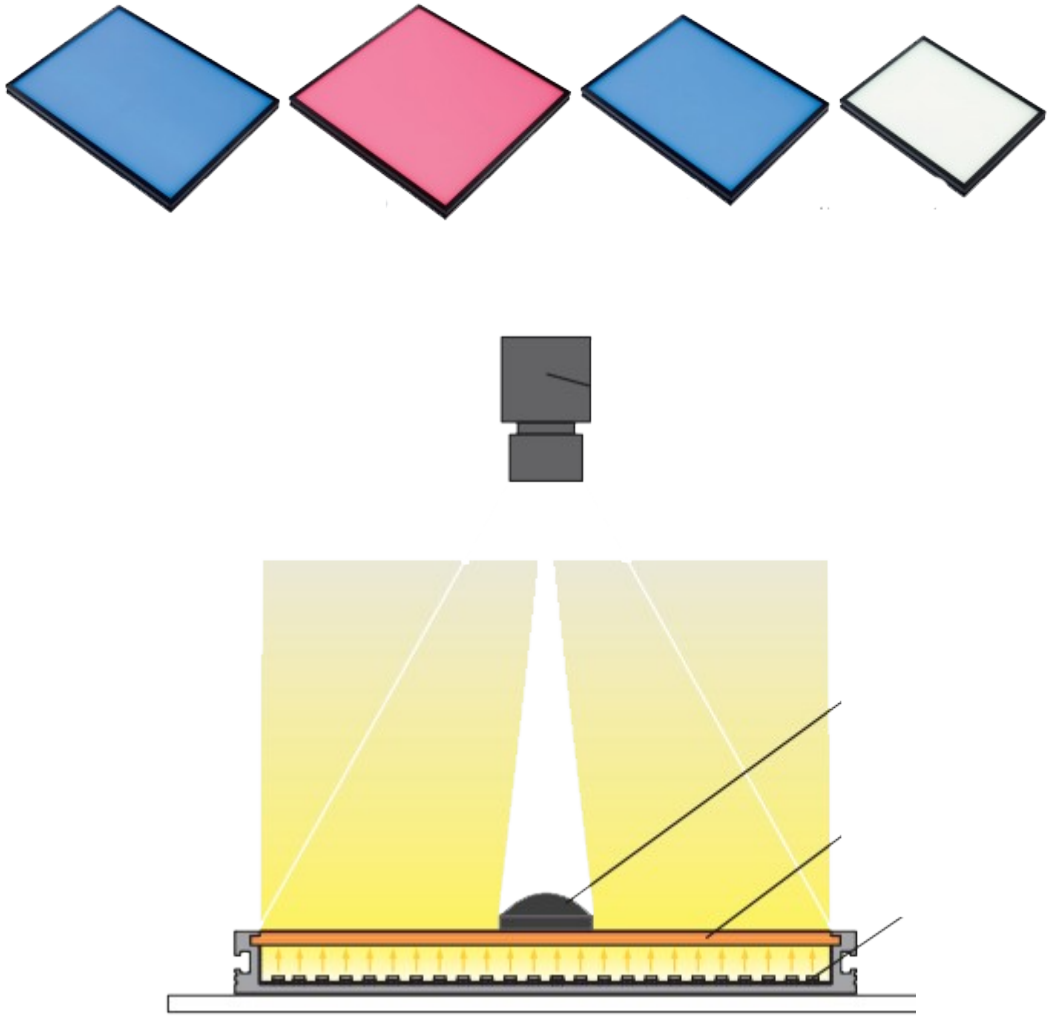


条光

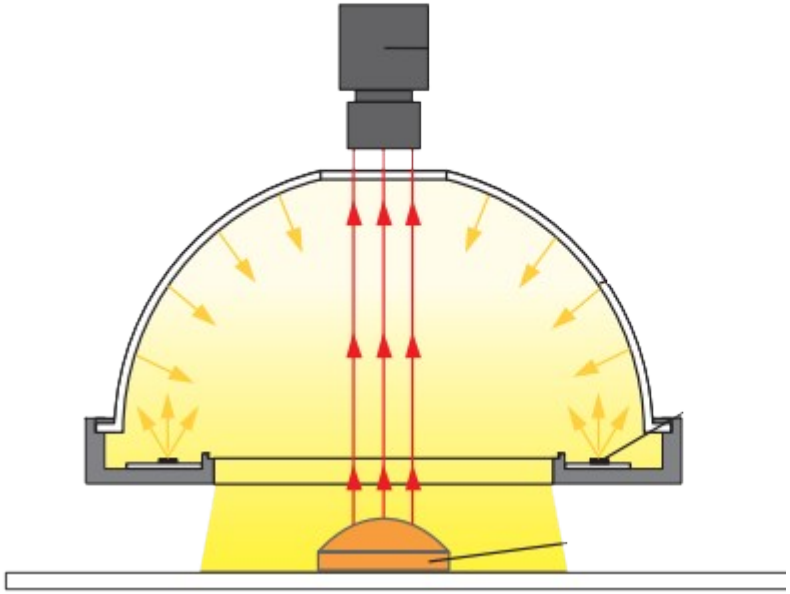




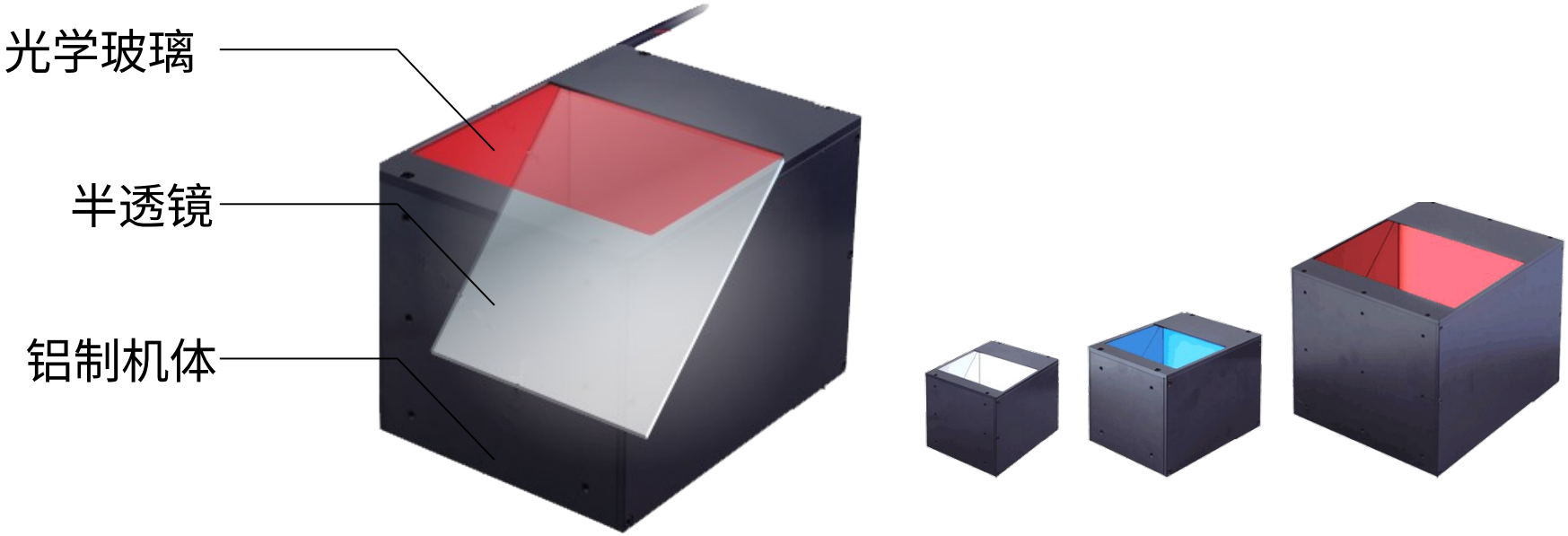
面光



弧顶光



同轴光



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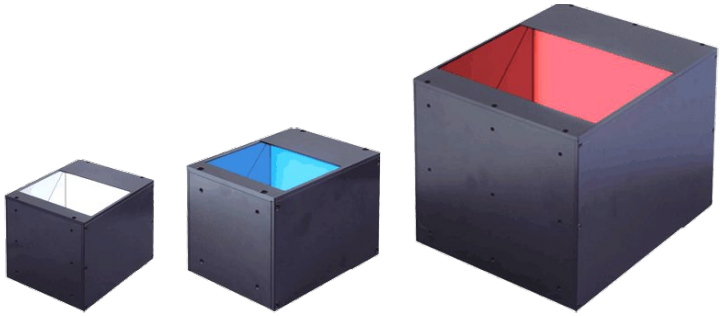
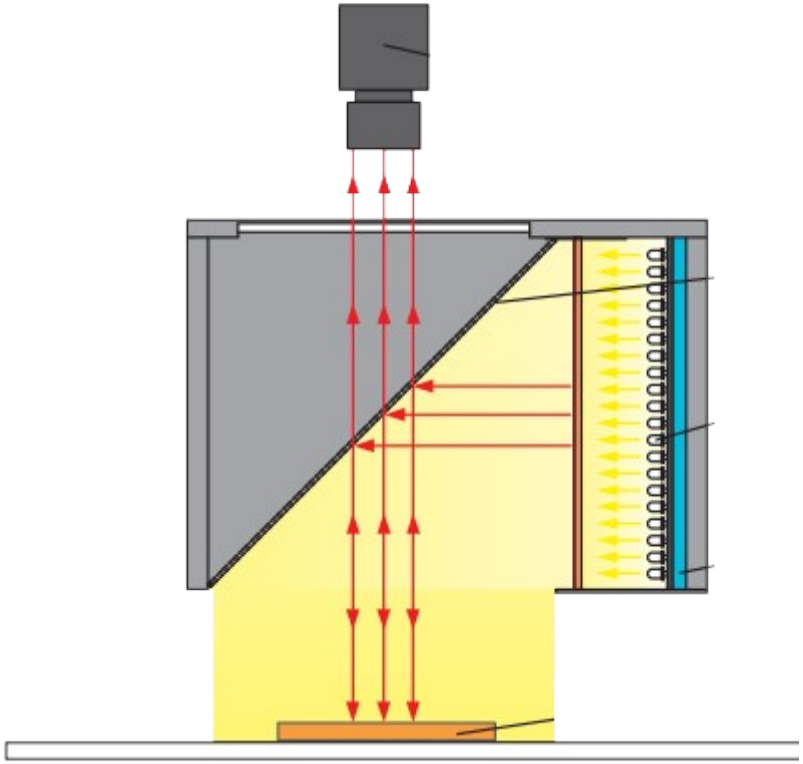
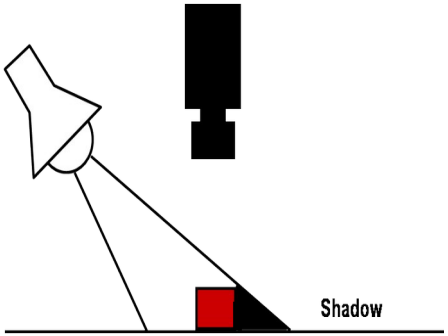




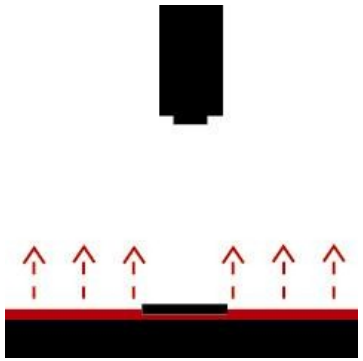
Overview – 光源

Type	响应速度 Response Speed	价格 Cost	亮度 Intensity	寿命 Life	发热 Heat
卤素灯 Halogen	Poor	Fair	Good	Poor	Poor
白炽灯 Incandescent	Poor	Good	Fair	Poor	Poor
荧光灯 Fluorescent	Poor	Good	Poor	Fair	Fair
激光 Laser	Good	Poor	Good	Good	Good
二极管 LED	Good	Fair	Fair	Good	Good
疝气灯 Xenon	Good	Poor	Good	Poor	Poor

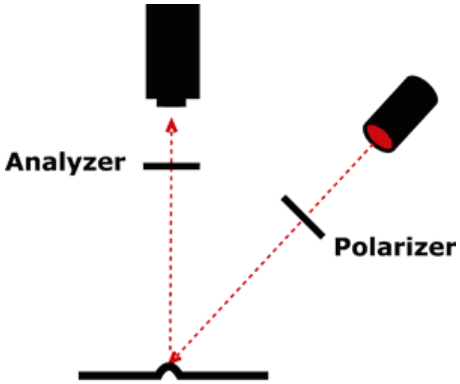
照明技术



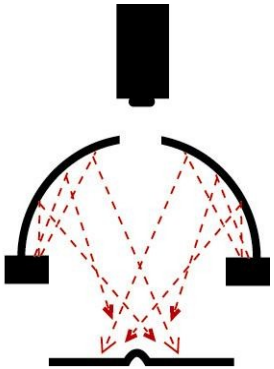
Direct (or Front)



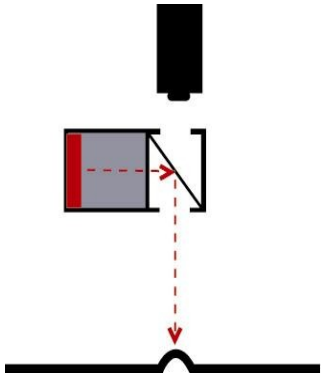
背光源



偏振光

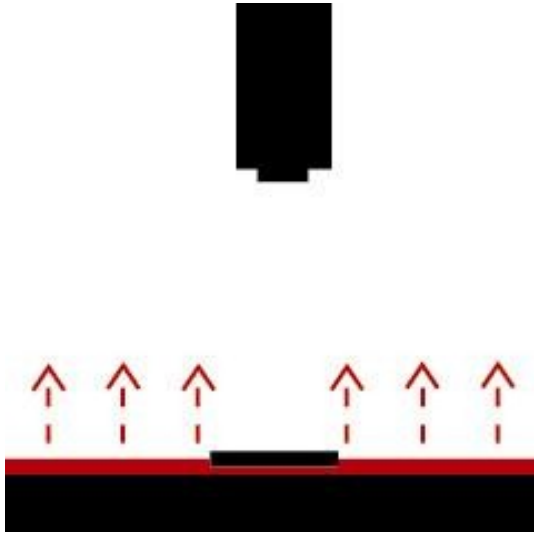


非同轴漫射光



同轴漫射光

背景光源



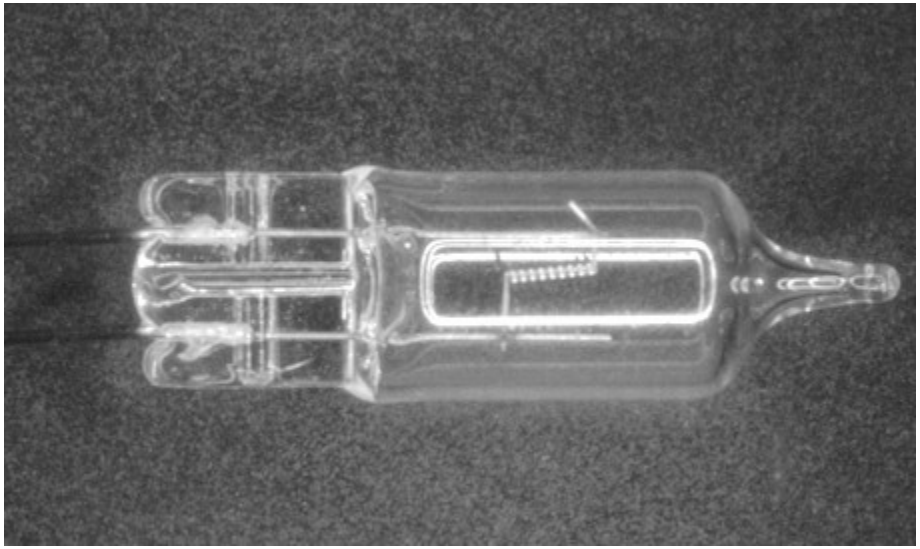
背光

提供最佳对比度 .

- 零部件的轮廓和边线检测 .

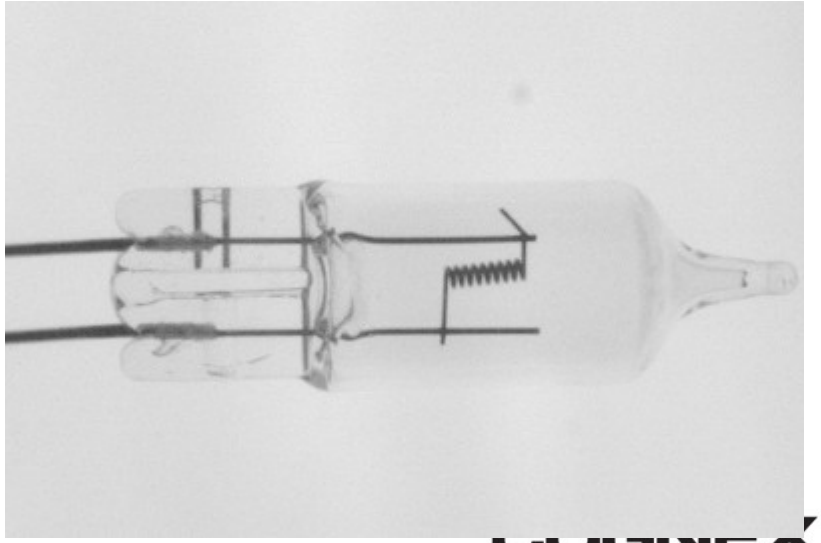
丢失的光源非常少 .

背光源应用



环形光源

背光源

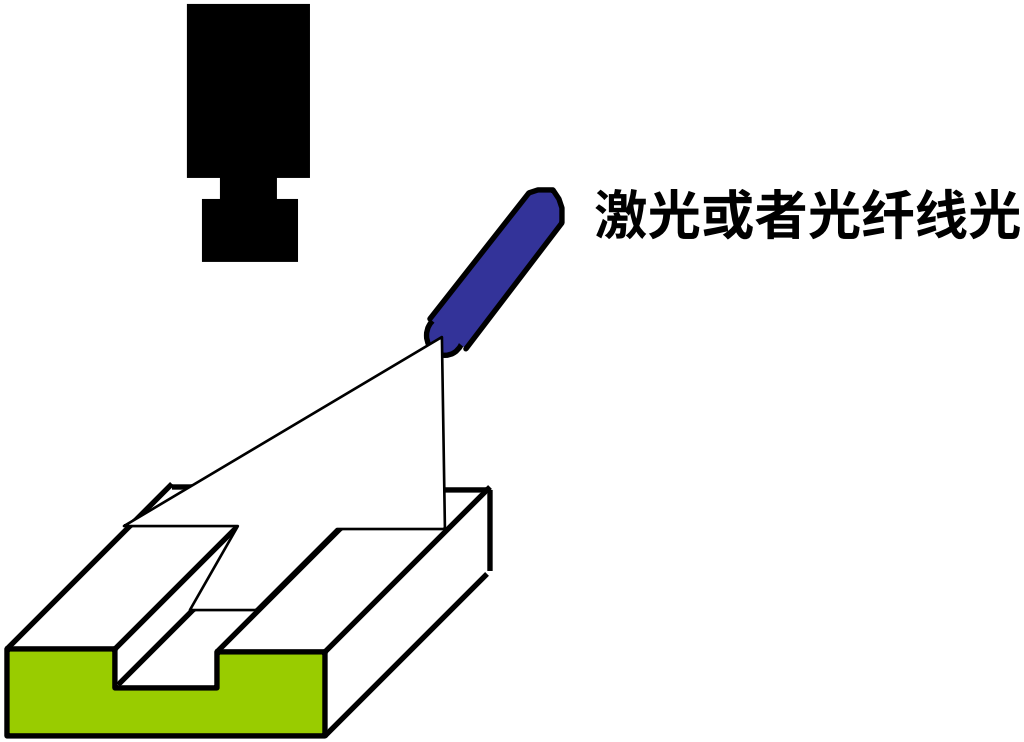


背光源应用

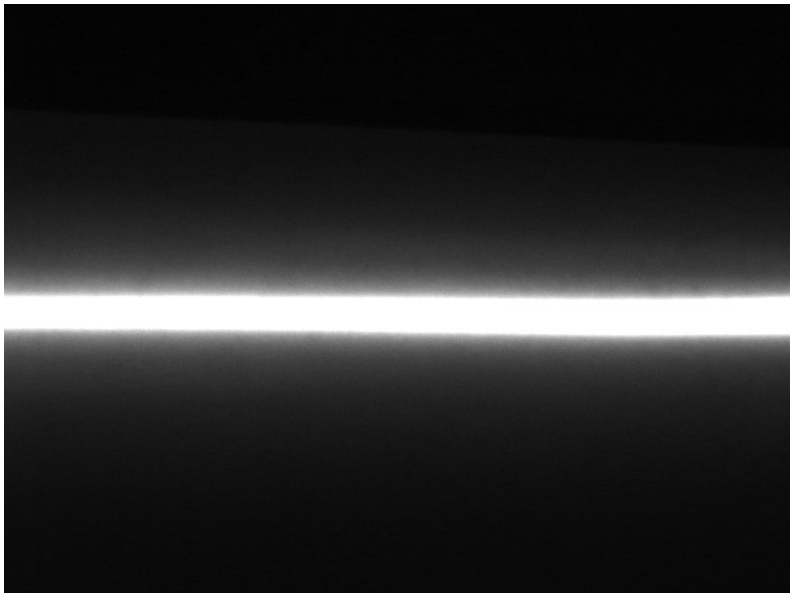
Ring Light

Back Light

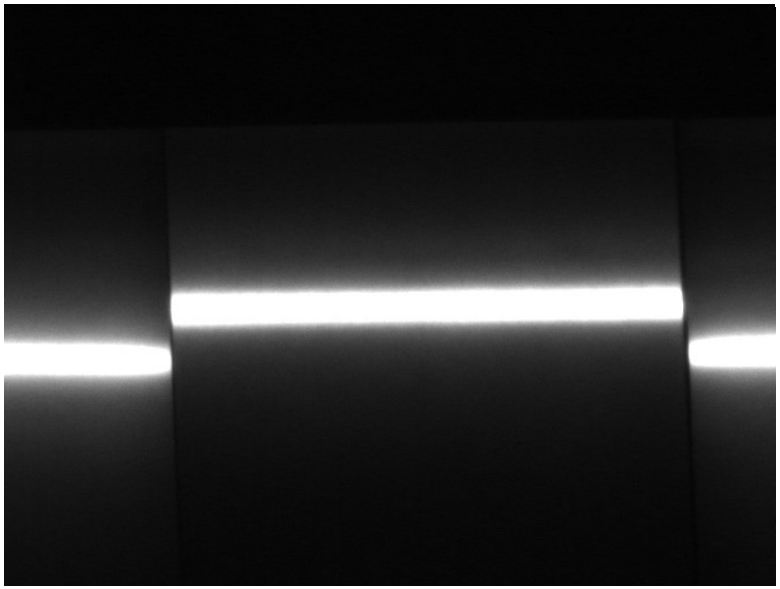
结构光



激光应用

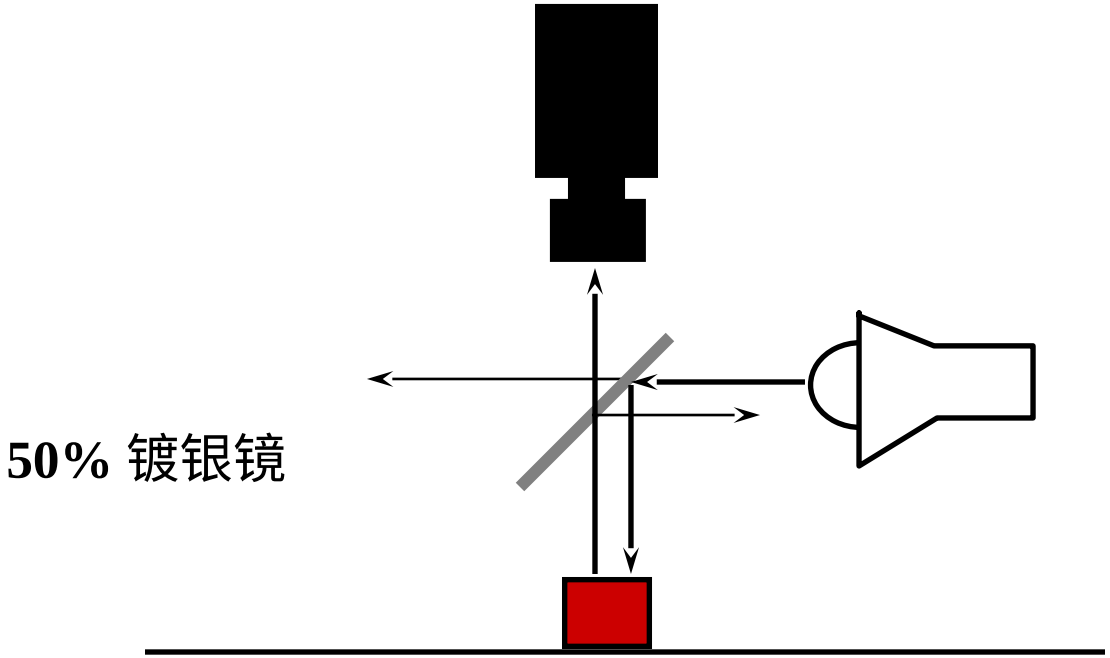


线性光源

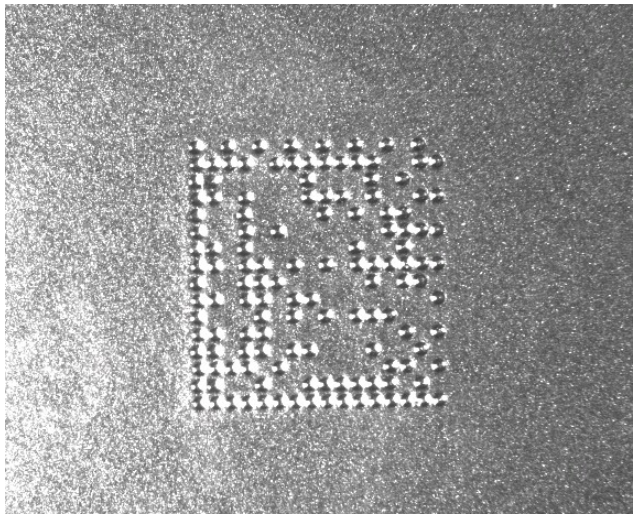


高度上有变化

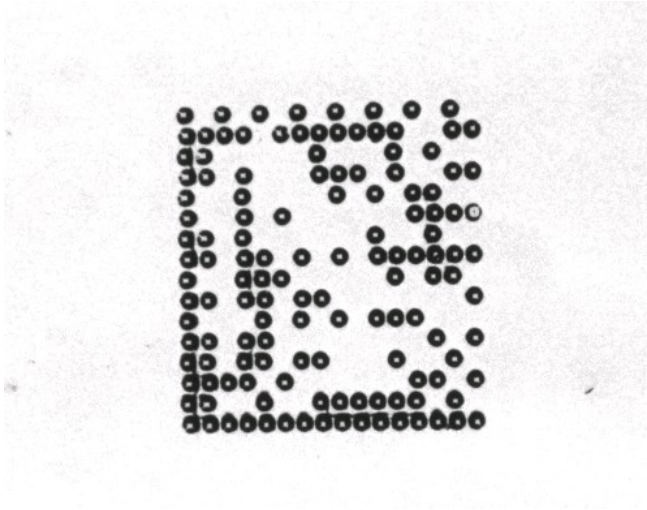
同轴光



同轴光应用

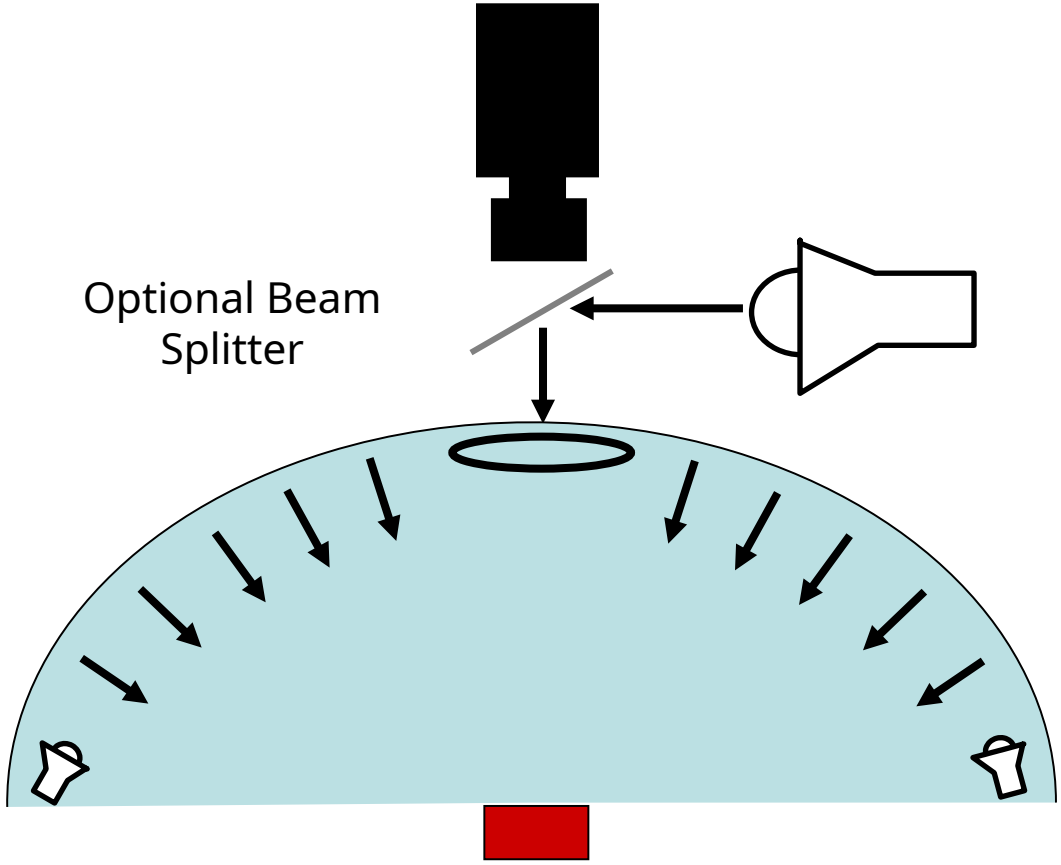


环境光

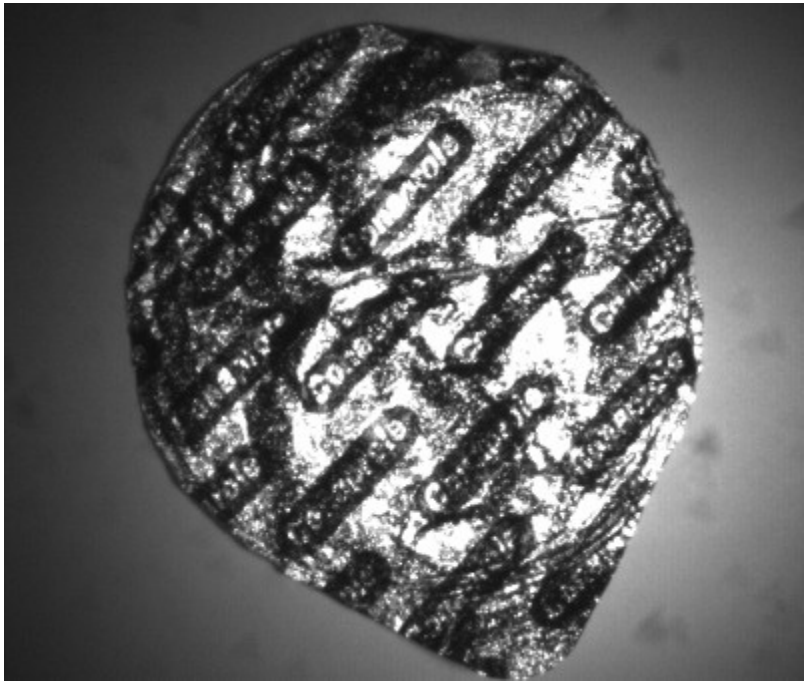


同轴光

非同轴漫射光 (Dome)



Dome 光应用

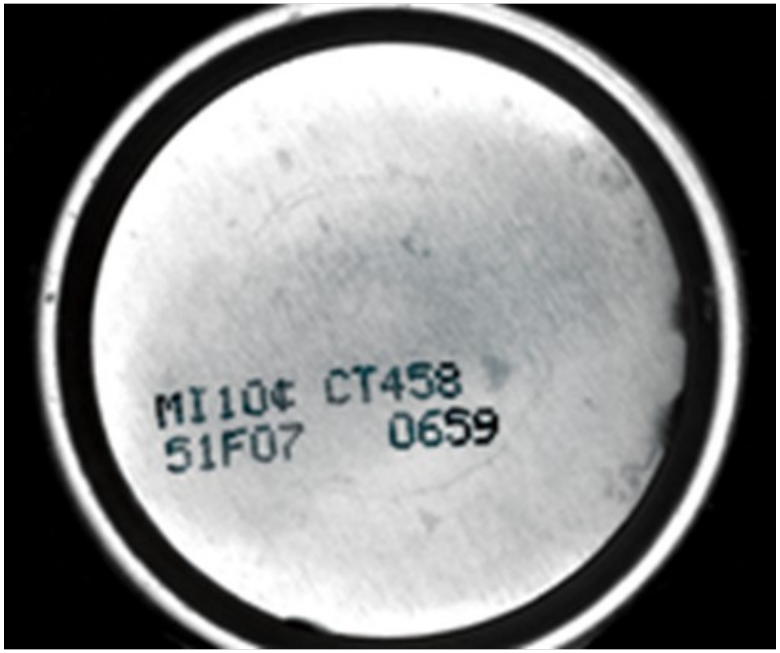


用直向型前光源的铝铂密封

用 CDI 的铝铂密封

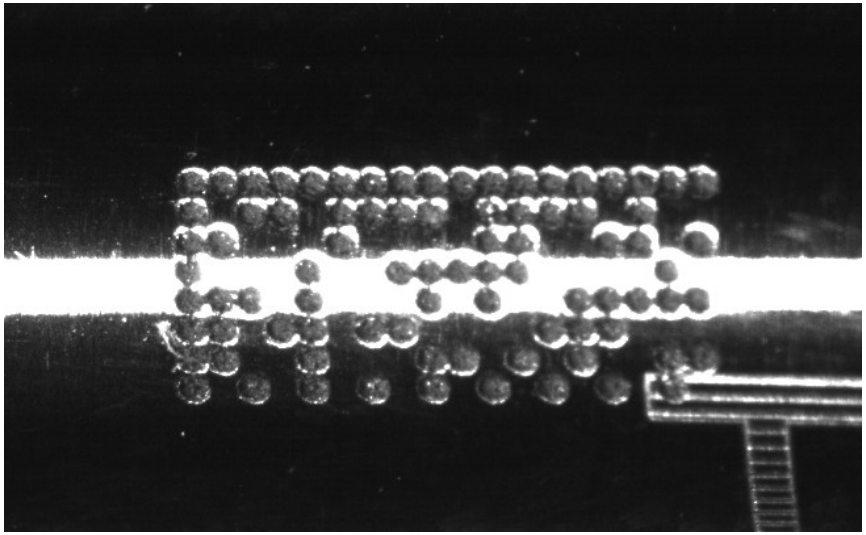
Dome 光应用

环境光

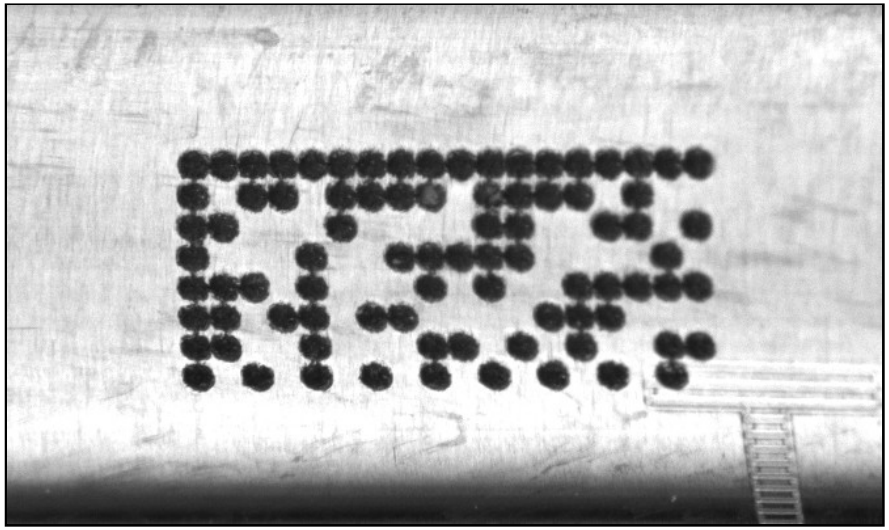


Dome 光

Dome 光应用



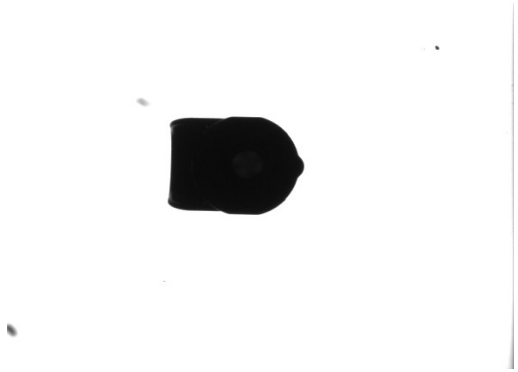
环形光



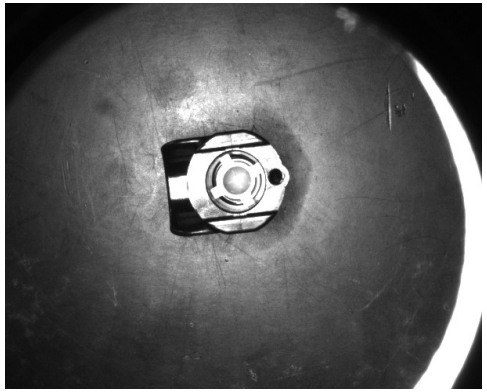
非同轴漫射光 (Dome)

Review of Techniques

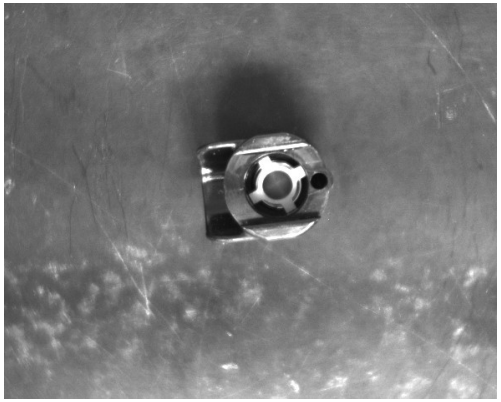
Front Light



Back Light

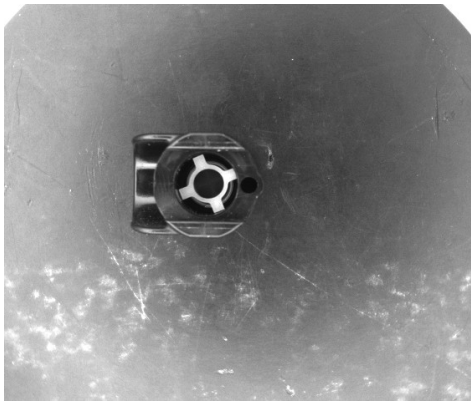


Ring Light



Ambient Light

Diffuse On-Axis (DOAL)

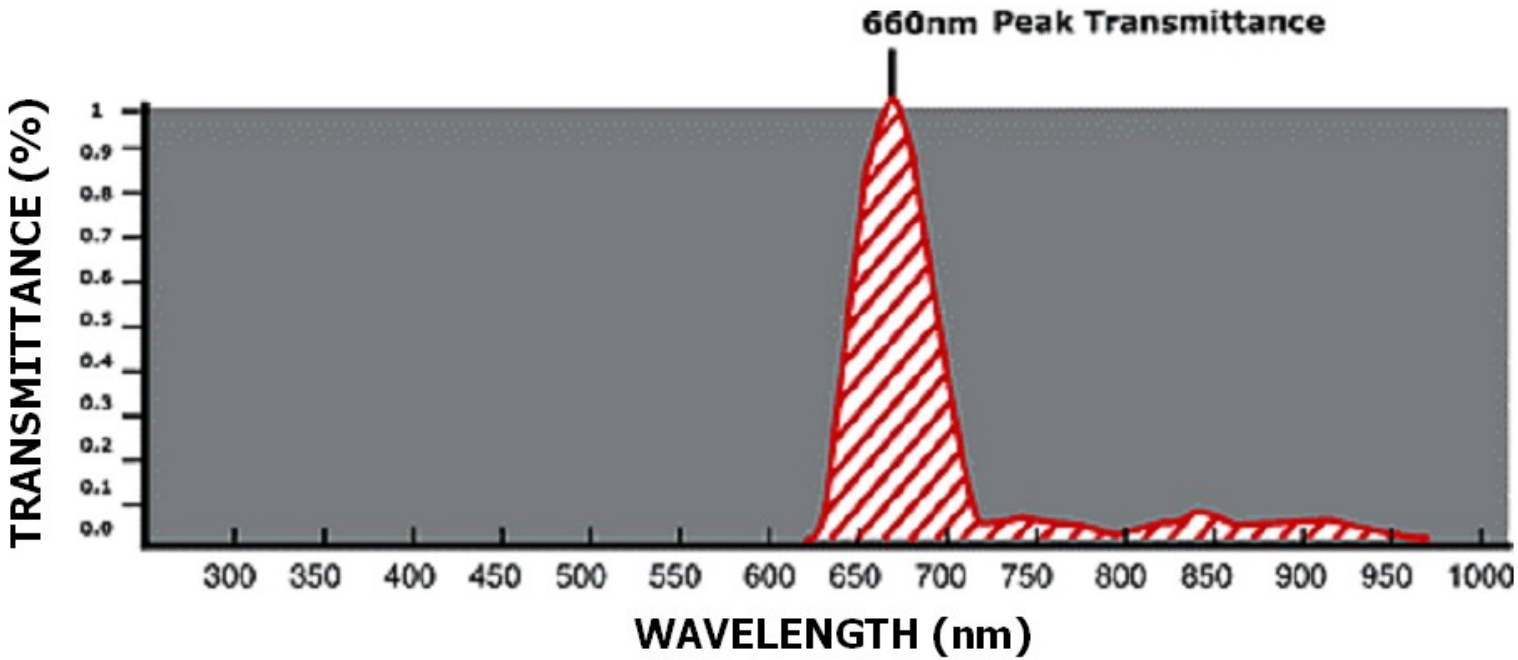


Diffuse Off-Axis

Filters and Colored Light

COGNEX

过滤器

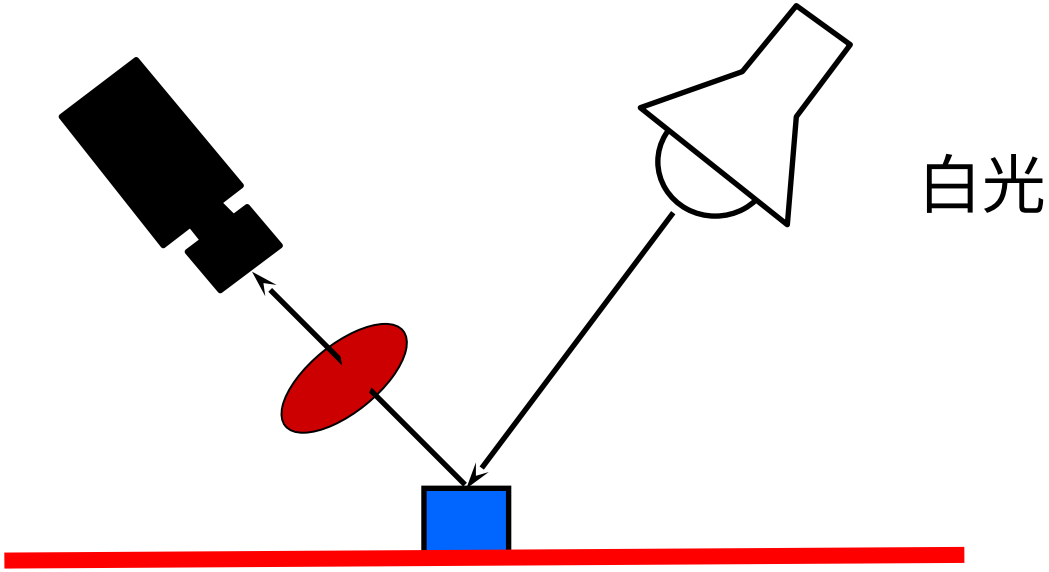


可以选择所通过的光 .

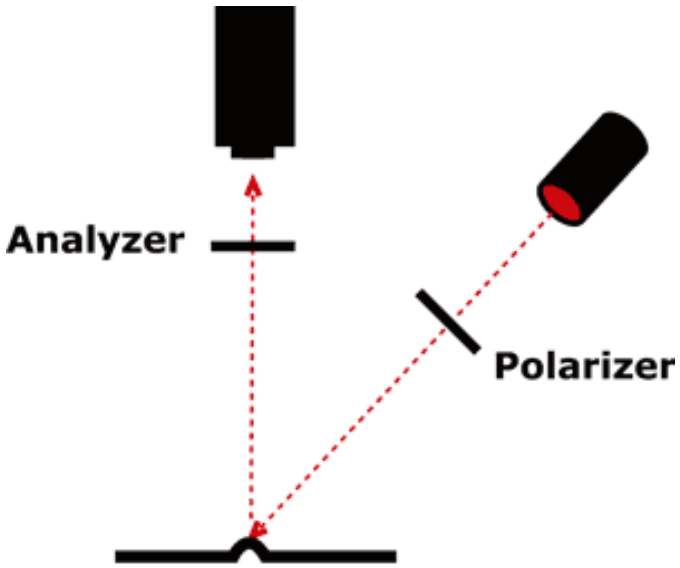
- 低通 , 高通

有效减少环境光的干扰 .

滤镜



偏振光

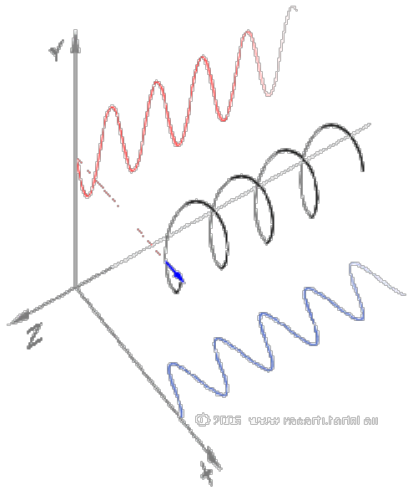


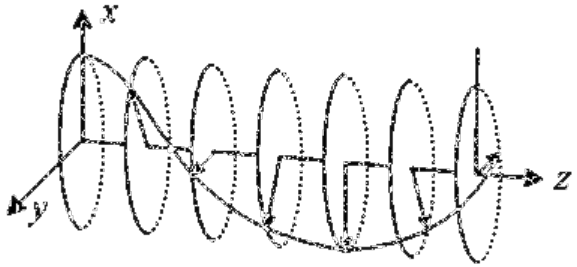
用于减少眩光或者是镜面反射 .

光线经偏振片过改变传播方向 .

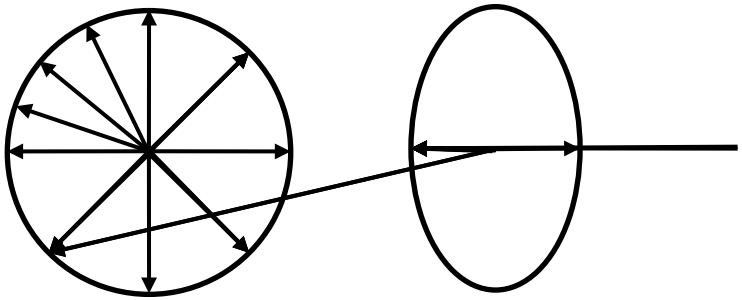
镜头前的偏振片配合使用 .

偏振光





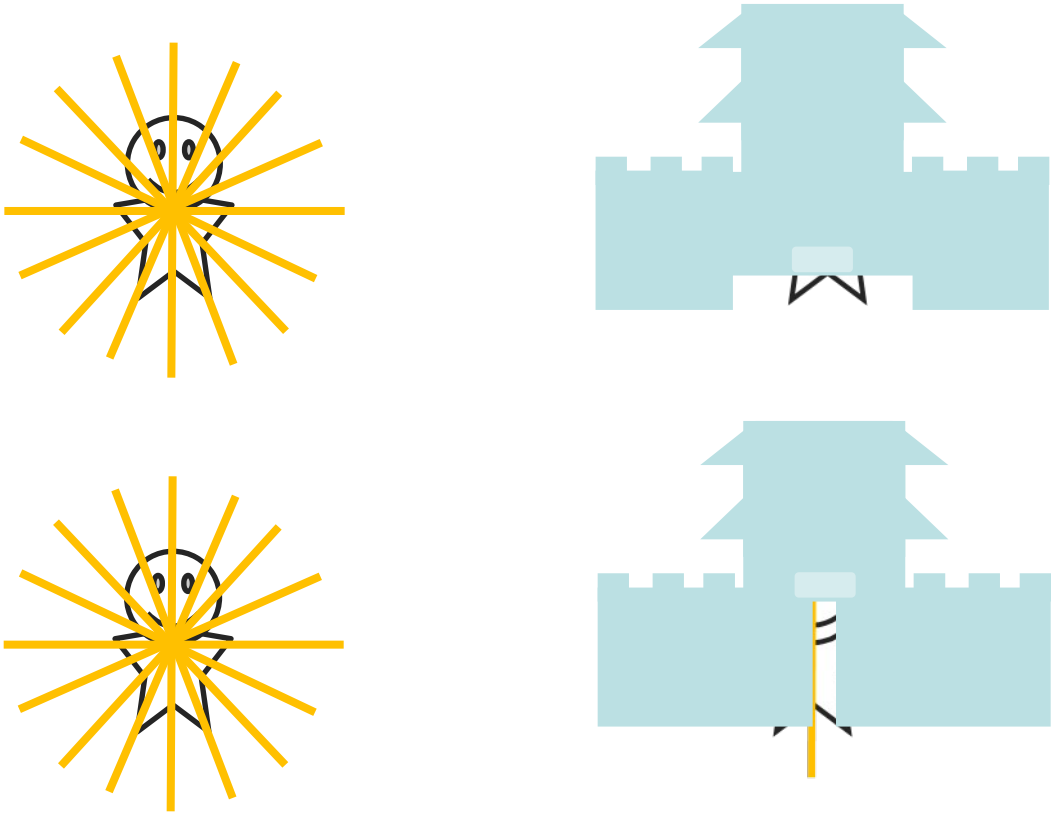
偏振态



圆偏振　　椭圆偏振　　线偏振

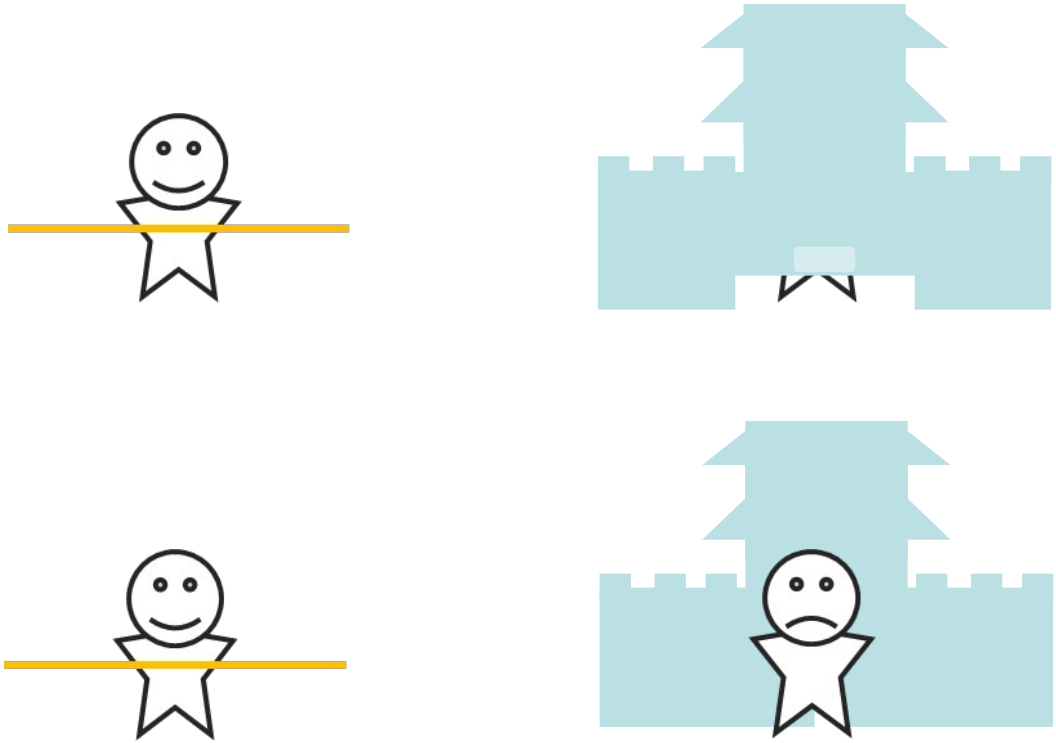
COGNEX

偏振光



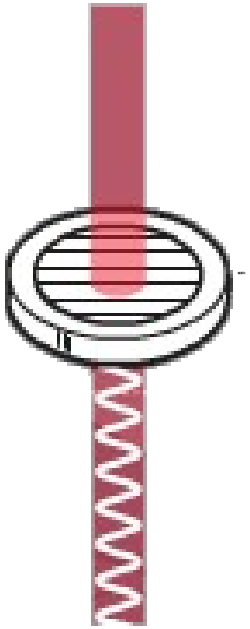
COGNEX

偏振光

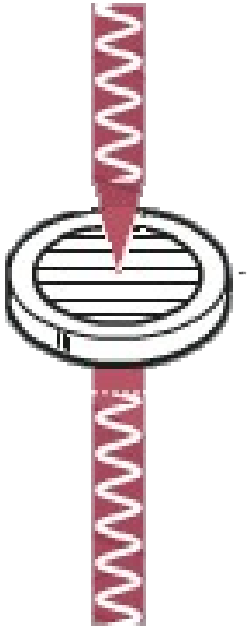


COGNEX

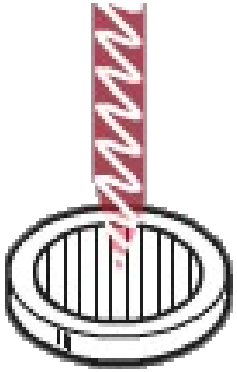
偏振光



线偏振光的产生



偏振态与线偏振光一致

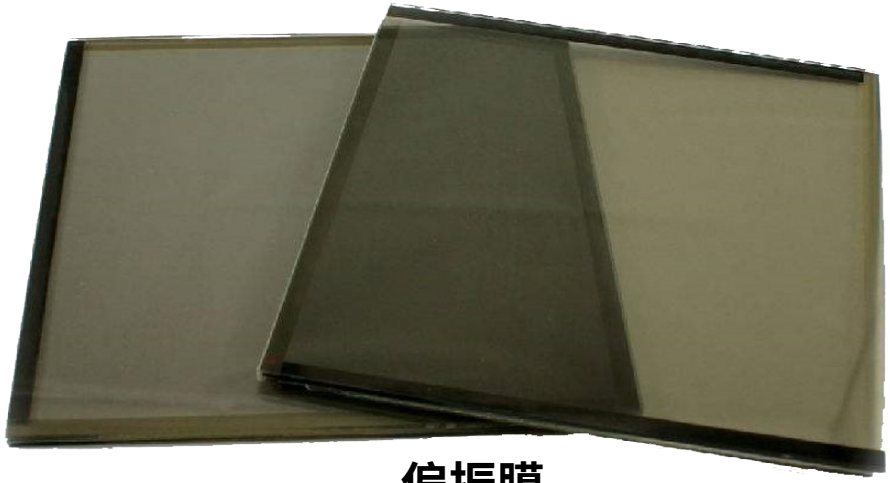


偏振态与线偏振光垂直

COGNEX

偏振光

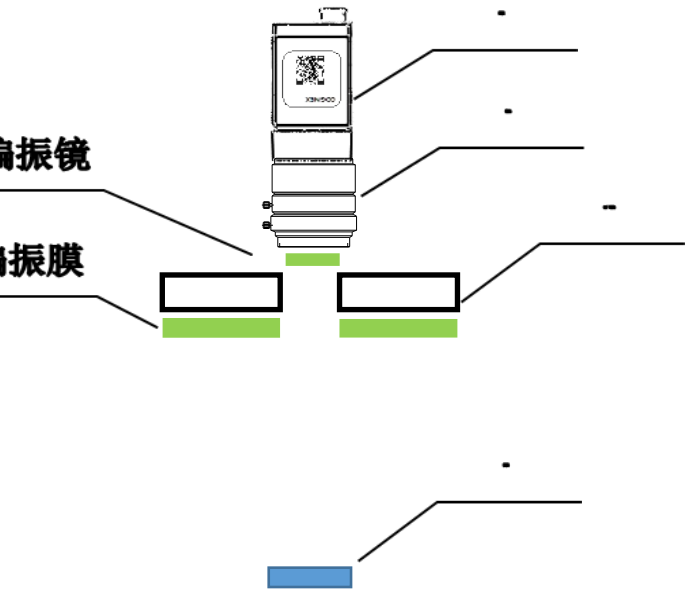
偏振镜



偏振膜

COGNEX

偏振光



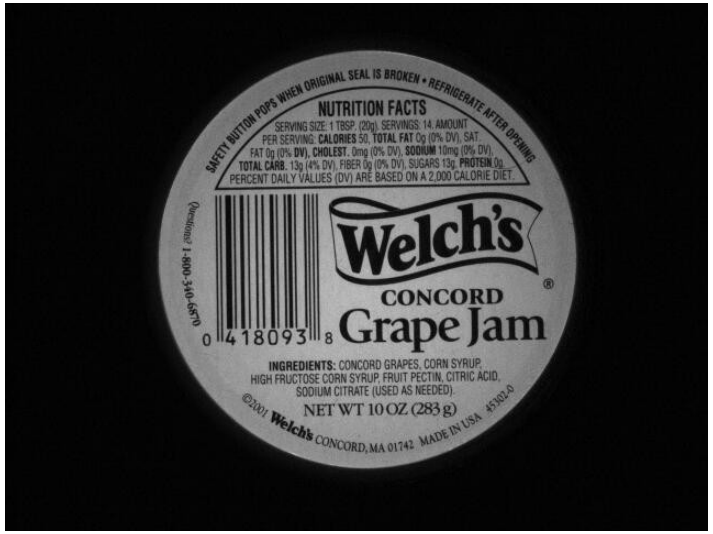
未加偏振镜、偏振片

加偏振镜、偏振片

COGNEX

偏振光应用

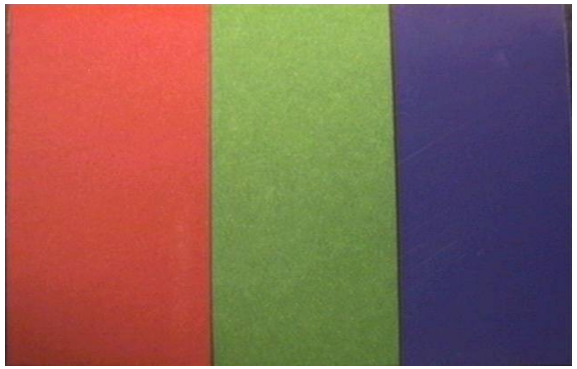
无偏振环形光

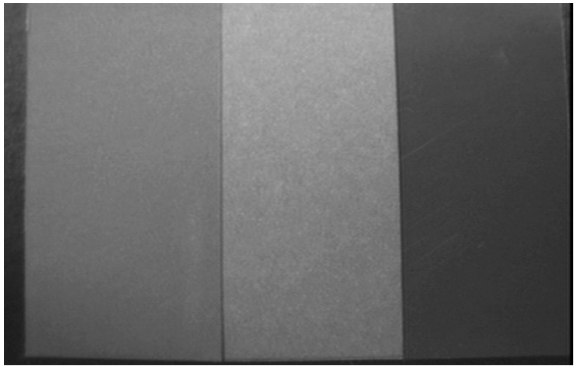


带偏振环形光

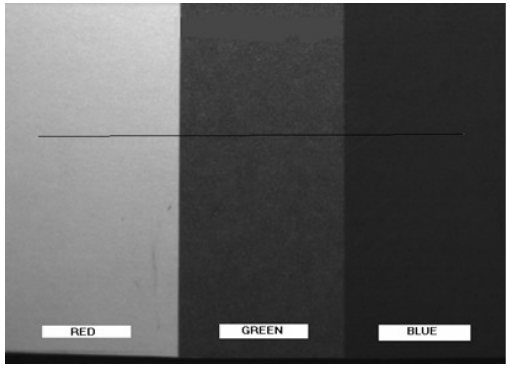
COGNEX

彩色应用

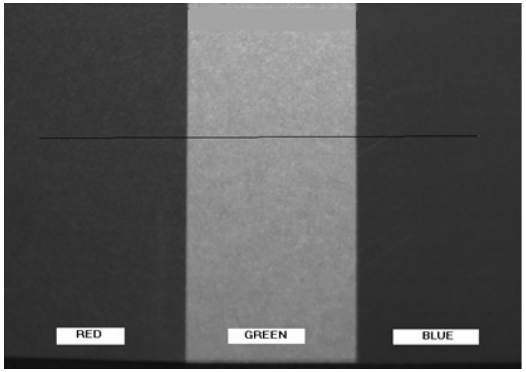




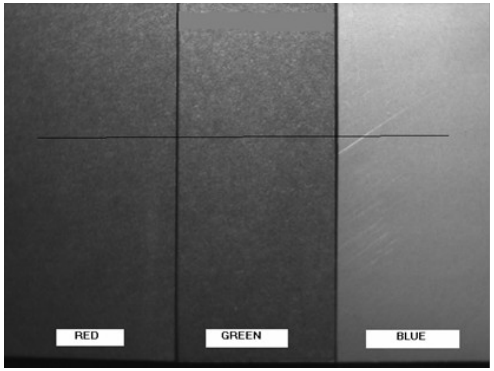
白光



红光

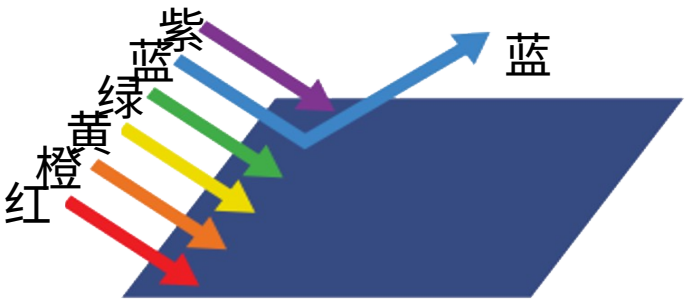


绿光



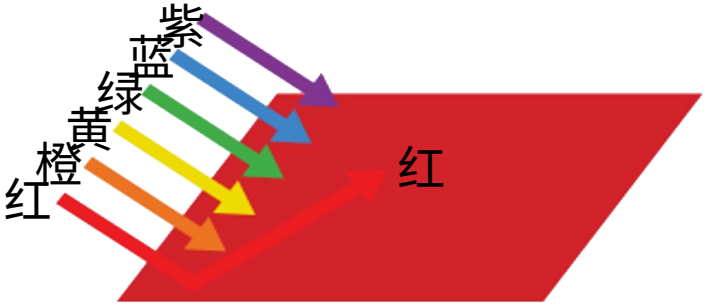
蓝光

互补、相近色



蓝色物体

白色物体



红色物体

黑色物体

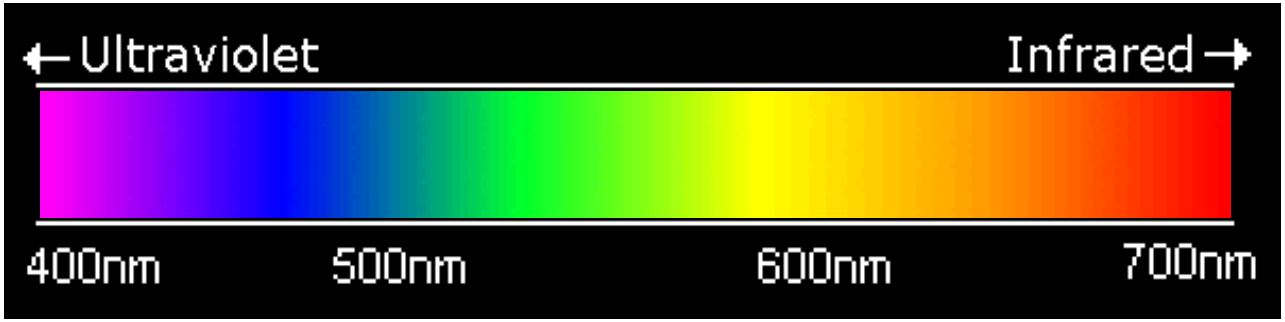
彩色光源应用

环境光

蓝光

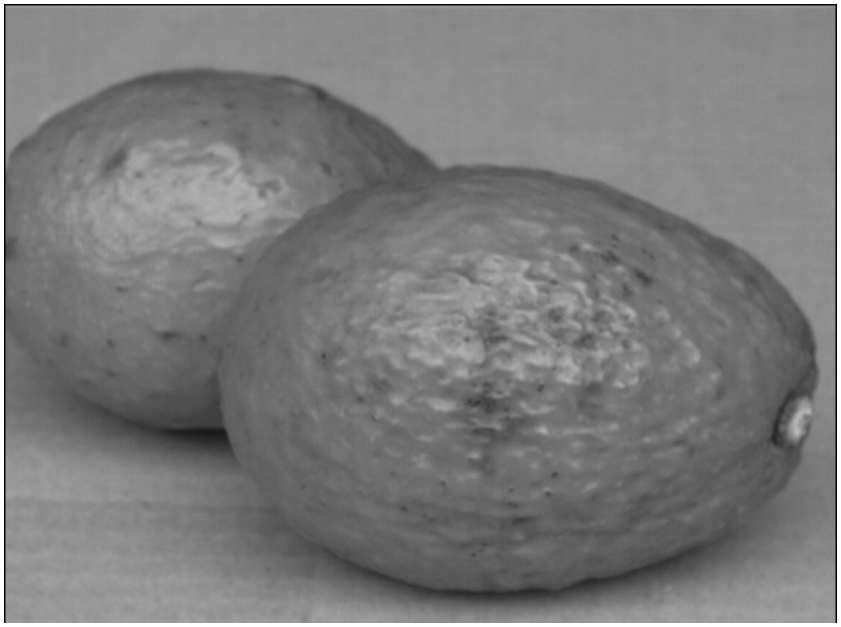
- Date information is accentuated with blue light
  - 蓝色变淡
  - 红色加深

红外光

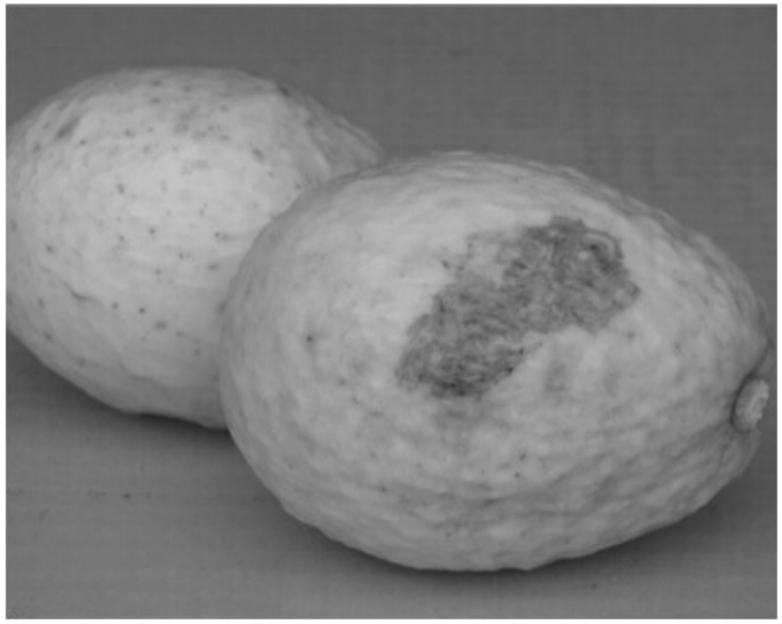


- IR 光人的肉眼不可见
- 操作人员不知道使用了红外光源
- 红外光的穿透性能强

红外光应用



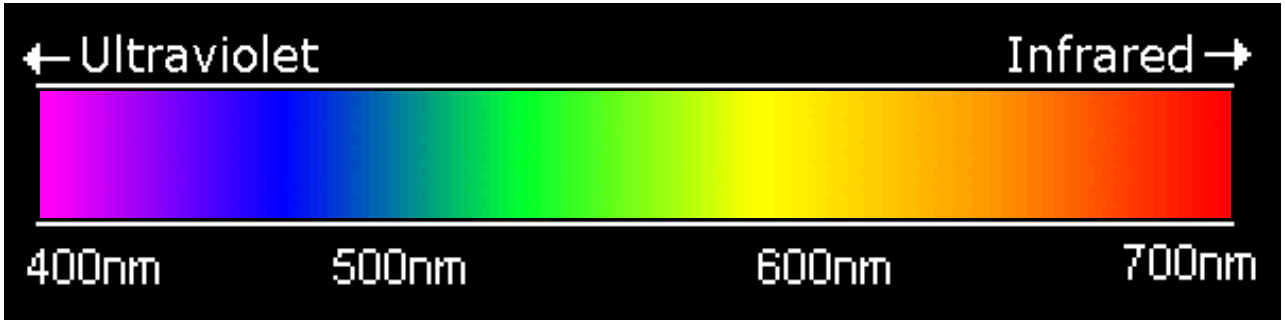
环境光



红外光及滤镜
(IMIF-BP850)

* 多数 IR 和 UV 的应用会用到滤镜

紫外光



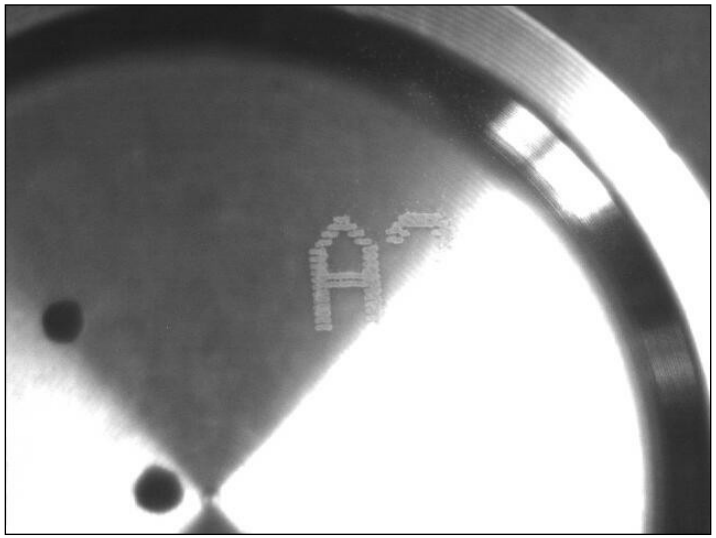
很多物体在紫外光的照射下会发出荧光 .

- 墨水 , 标签 , 胶水

直接用 UV (or Near-UV) 光源照射到物体上 , 物体将发出荧光 .

可使用过滤器将 UV 光过滤掉 , 仅允许荧光波长的光源进入相机 .

紫外光应用



不滤光

高通 (UV) 滤光器

使用 UV 背景光源 , 墨水字发出荧光 .

Question 1

- 什么光源最适合用于金属圆柱物体表面检测?
  - A. 暗场 Dark Field
  - B. 背光 Back Light
  - C. 亮场 Bright Field
  - D. Dome (Cloudy Day Illumination/Dome)

D

Question 2

- 下图中仅需检测蓝色字符，请问使用什么光源?

A. 红光
B. 绿光
C. 蓝光
D. 红外光

A

COGNEX